Generative AI Declaration

Tool: Claude (Anthropic), used through Claude Code.
Purpose: Double checking and debugging of code to ensure that experiments are runnable
Verification:

- Syntax & Import Validation: Performed a comprehensive codebase scan using Claude Code to verify that all dependencies are accounted for, relative file paths are correct, and no unresolved imports or syntax errors exist

Question B1

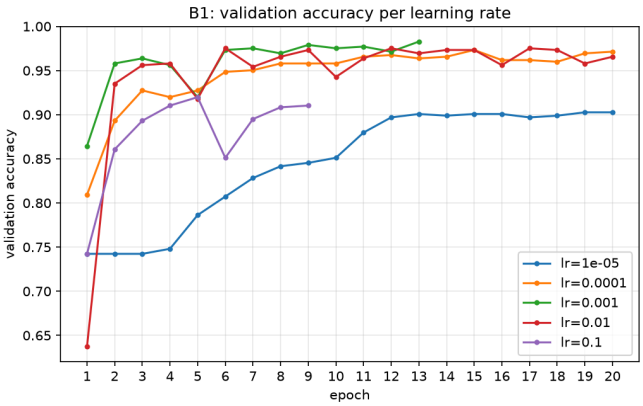


Figure 5 (a) Validation accuracy for 5 different learning rates

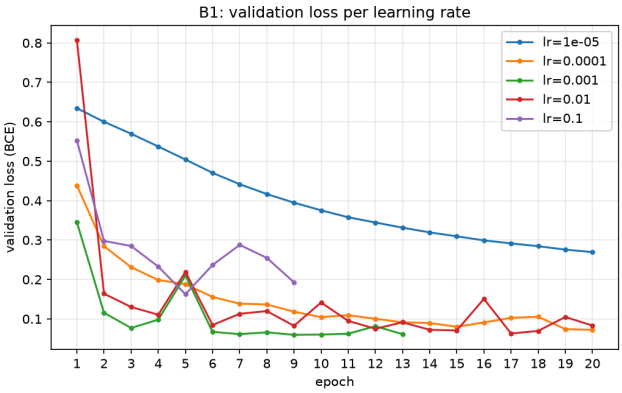


Figure 5(b) Validation loss for 5 different learning rates

learning rate	best epoch	val loss	val error	sensitivity	specificity
1e-5	20	0.2695	0.0973	0.9614	0.7333
1e-4	20	0.0725	0.0286	0.9794	0.9481
1e-3	9	0.0597	0.0210	0.9897	0.9481
1e-2	17	0.0631	0.0248	0.9743	0.9778
1e-1	5	0.1626	0.0802	0.9846	0.7333

Figure 5(c) Best results for 5 different learning rates

Optimal learning rate: 1e - 3

1e-3 was selected as the learning rate because it provided the best balance between convergence speed and training stability. Smaller learning rates (1e-5,1e-4) reduced the loss more slowly and the 20 epoch is insufficient to observe full convergence. Larger learning rates (1e-2, 1e-1) showed increasingly unstable training behaviour, with more fluctuations observed and possibly triggering early stopping, suggesting learning rate is becoming too large, causing the optimiser to overshoot the optimal region rather than converge smoothly. At 1e-3, the model reached a low training loss quickly without the severe fluctuations observed at higher learning rates. Furthermore, 1e-3 achieved the lowest validation error (2.10%) and the

best validation loss (0.0597), sensitivity (0.990), and one of the best specificity (0.9481, tied with 1e-4). The validation accuracy was also the highest among the tested learning rates.

The validation set has an imbalance set of pneumonia and healthy images. If only accuracy is used, it provides a single overall measure, but with imbalanced classes or unequal error costs, it can hide poor performance on the minority class. Sensitivity and specificity show how well the model detects sick patients and avoids falsely flagging healthy patients which is detrimental, thereby providing a more complete assessment of its reliability.

Test Result:

Accuracy	Sensitivity	Specificity
0.8125	1.0000	0.5000

Question B2

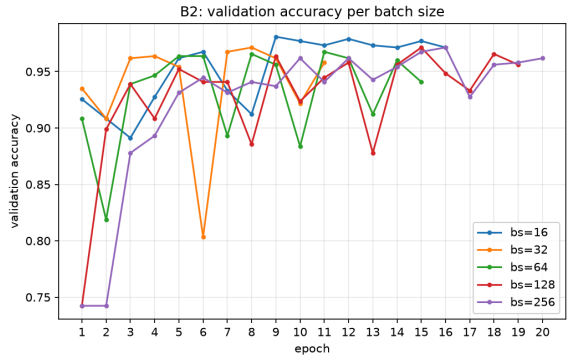


Figure 6 (a) Validation accuracy for 5 batch sizes

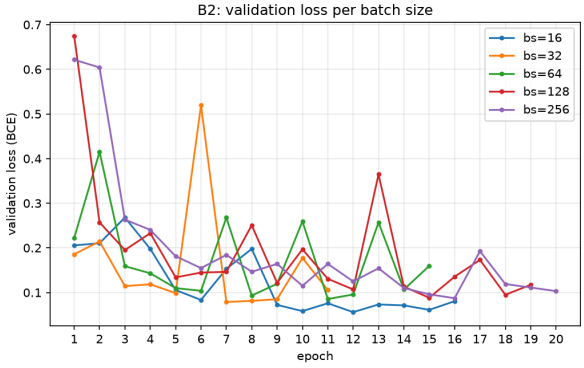


Figure 6 (b) Validation loss for 5 batch sizes

batch size	best epoch	val loss	val error	sensitivity	specificity
16	**12**	**0.0560**	**0.0210**	**0.9871**	**0.9556**
32	7	0.0788	0.0324	0.9666	0.9704
64	11	0.0855	0.0324	0.9769	0.9407
128	15	0.0882	0.0286	0.9974	0.8963
256	16	0.0872	0.0286	0.9820	0.9407

Figure 6 (c) Best results for all batch sizes

Optimal batch size: 16

It has the lowest validation classification error (0.0210) and also the lowest validation loss (0.0560), so the choice does not depend on one metric. Smaller batches make more parameter updates per epoch, and their noisier gradients are associated with better generalisation. bs=16 is the most stable after epoch 9, whereas the other batch sizes show large oscillations.

Larger batches converge more slowly within the 20-epoch budget. bs=128 and 256 predicted every image as pneumonia at epoch 1 (accuracy 0.742), so their large early gains reflect recovery from a poor start rather than fast learning. bs=256 reached its best epoch only at 16, suggesting it had not fully converged. Although larger batches give more accurate BatchNorm statistics, the far fewer updates outweigh this within a fixed epoch budget.

bs=16 also gives the best sensitivity–specificity balance. It beats bs=64 on both (0.987/0.956 vs 0.977/0.941) and bs=256 on both. bs=128 has higher sensitivity (0.997) but much lower specificity (0.896). bs=32 has higher specificity (0.970) but lower sensitivity (0.967); since missing a pneumonia patient is the more serious error, bs=16 is preferred. Differences are small and based on a single run each.

Test Result:

Accuracy	Sensitivity	Specificity
0.8782	0.9949	0.6838

Question B3

architecture	accuracy	sensitivity	specificity	error	parameters	FLOPs
AlexNet-like (B1)	0.8125	1.000	0.5000	0.1875	99,569	39,063,616
VGG-like (B2)	0.8782	0.9949	0.6838	0.1218	18,185	19,464,256
ResNet-like (B3)	0.9119	0.9846	0.7906	0.0881	77,169	25,297,024

Figure 7 Performance of AlexNet-like, VGG-like, ResNet-like

AlexNet-like (B1)

Pros: The architecture is simple and shallow. The 5×5 kernels capture large regions early while the 4×4 average pool keeps some spatial layout for the classifier and dropout enabled regularises the fully connected (FC) layers.

Cons: It has the most parameters and the most compute/FLOPs. 68% of its parameters (67,713) are in the FC layers, (mainly 1024→64), which raises the risk of overfitting. The 5×5 conv on 32×32 maps accounts for about 67% of its FLOPs. It has the lowest accuracy and specificity. Although it labels every pneumonia case correctly, it misclassifies half of the normal patients, which shows a strong bias towards predicting pneumonia, making the model weak.

VGG-like (B2)

Pros*:* It has the fewest parameters (5.5× fewer than B1) and the fewest FLOPs (about half), yet it outperforms B1 on test accuracy and specificity. Two 3×3 convs cover the same receptive field as a 5×5 with fewer weights and an extra non-linearity, and global average pooling removes almost all FC parameters.

Cons: A plain stack with no shortcuts, so deeper versions would become harder to optimise as gradients weaken through many layers. Its first two convs run at the full 64×64 resolution, costing 27% of its FLOPs for only 8 channels. It ends with only 32 channels, which limits capacity, and global average pooling to 1×1 discards spatial layout before the classifier. Its specificity (0.684) and accuracy (0.878) remain below ResNet-like's (0.791 and 0.912)

ResNet-like (B3)

Pros: The highest test accuracy, specificity and the lowest error. Shortcuts give gradients a direct path, which eases optimisation and allows greater depth. Early max pooling and stride-2 convs keep FLOPs only 30% above VGG-like despite 4.2× the parameters.

Cons: It has 77,169 parameters, 74.8% of them in the final 64-channel block. The 1×1 projection shortcuts add a small overhead (2,752 parameters, 2.1% of FLOPs). It is also the most complex of the three to implement. the lowest sensitivity (misses 6 of 390 pneumonia cases vs 2 for B2 and 0 for B1), trading a few missed cases for correctly clearing more healthy patients.

Overall, ResNet-like gives the best performance for a moderate cost, VGG-like the best efficiency, and AlexNet-like is the least efficient with the weakest results.

Question B4

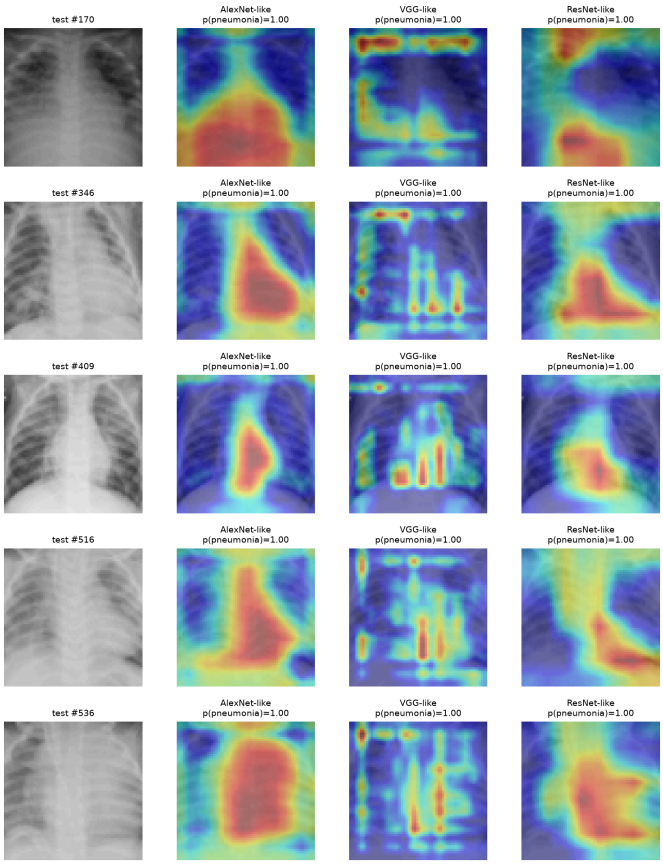


Figure 8 Grad-Cam Heatmap of AlexNet-like, VGG-like, ResNet-like on 5 random samples

AlexNet-like: large, smooth activation regions centred on the middle and lower chest. In #170 and #536 they cover most of the lower thorax, including the heart and mediastinum.

VGG-like: fragmented, point-like and striped activations spread across the lung fields, with a few sharp hotspots near the image edges. The hardest to interpret anatomically.

ResNet-like: coarser maps, because of its 8×8 final resolution. They are compact regions over the lower-central chest and the adjacent lower lung zones, for example the lower right of #516.

All three models placed most of their attention on the central and lower chest. The central strip covers 27% of the image but receives 37–44% of heatmap weight. AlexNet-like is the most centrally focused, giving the lung fields less weight than their area (36% vs 49%). VGG-like weights the lung fields in proportion to their area (49%), but rarely forms coherent regions that could correspond to a patch of consolidation. All three place 57–62% of their weight on the lower half, which fits lower-lobe pneumonia, although the heart and diaphragm also lie there. The highly activated regions are only marginally brighter

than the image average (0.04–0.10 SD), so there is little evidence that the models specifically target white and cloudy areas. AlexNet-like and ResNet-like produce the most similar maps (correlation 0.59), while VGG-like is the least similar to either (0.37 and 0.44).

Overall, Grad-CAM did not clearly isolate the white, cloudy patches in any of the three architectures. All three rely substantially on the central chest, which suggests they may partly use global cues such as the heart and mediastinum rather than localised opacities. ResNet-like, the best performer on the test set, gives the most coherent lower-chest regions, but they are not confined to the lungs.